Linear Algebra (MT-1004)

Homework#19

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Question 1

1. Apply the Gram-Schmidt process to the vectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, and $\mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.
2. Verify that the output vectors are orthogonal.
3. Normalize the orthogonal vectors to obtain an orthonormal basis.
4. **Python Task:** Write a function that implements the Gram-Schmidt process.

```python
def gram_schmidt(vectors):
    # Your code here
```